7.2 - Spanning Trees, MSTs, and MaxSTs

Problem Solving, Math 1000

Bryce A Christopherson

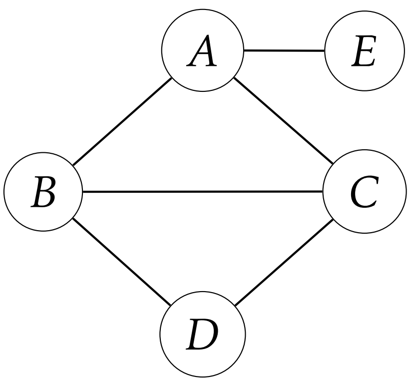
Definition

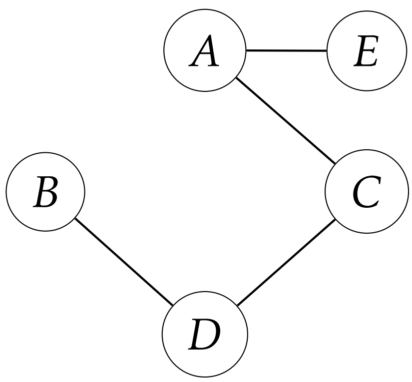
A spanning tree in a network is a subtree of the network which spans all of the vertices. That is to say, a spanning tree is a subgraph of the network which is a tree and which has an identical vertex set.

In a graph with no redundancy, there is only one spanning tree. In graphs with nonzero redundancy, there exists more than one spanning tree (often times, there exists many spanning trees).

Example

The graph on the right is a spanning tree of the graph on the left:





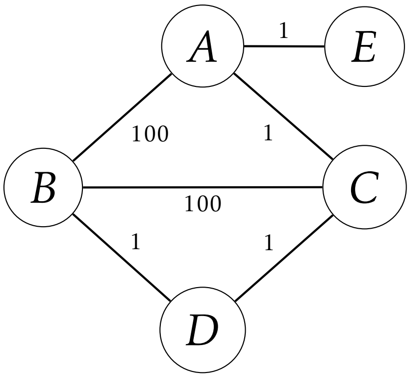
Definition

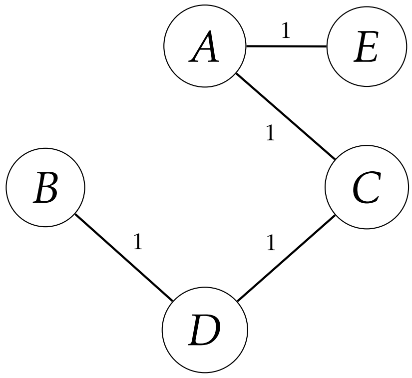
A minimum spanning tree (often denoted MST) is a spanning tree of a weighted network where the sum of the edge weights in the spanning tree is as small as possible.

In a weighted network with no redundancy, recall that there exists only one spanning tree. By defintion, this spanning tree must also then be minimal. In a network with nonzero redundancy, however, there are many spanning trees, and we have no guarantee that there will be only one minimal spanning tree for the network.

Example

The graph on the right is a minimal spanning tree for the graph on the left:





Definition

Similarly, a maximum spanning tree (often denoted MaxST) is a spanning tree of a weighted network where the sum of the edge weights in the spanning tree is as large as possible.

Example

The graph on the right is a maximal spanning tree for the graph on the left: